SUB-THEME: WASTE REDUCTION & CIRCULAR ECONOMY • ENVIRONMENTAL SOP STANDARDS

TEXCYCLE SYSTEM SPECIFICATION (VERSION 1.2)

System Engineering Design & Operational Specification for Intelligent Textile Waste Identification & Circular Governance in Micro-Garment Enterprises Based on On-Device Machine Learning (100% Offline Edge Intelligence)

Application Name	**TexCycle** (Textile + Cycle)	**Target Platforms**	Android (Mobile Native) & Web
Release Version	**1.2.0 (Advance Lens Edition)**	**Operating Mode**	100% Offline (Autonomous Edge Computing)
AI Engine	TensorFlow Lite (MobileNetV2 Transfer Learning)	**Database**	SQLite 3 (Embedded Local Storage)
Target Beneficiaries	Micro-Garments, Independent Tailors (1–10 Workers)	**Security Certification**	RSA 2048-bit Signature Schemes v1–v4 (Play Protect Safe)

1. SYSTEM IDENTITY & CORE DESIGN PHILOSOPHY

1.1 Industry Background & Urgency in Micro-Garments

Micro-scale garment workshops sustain local economies but face two acute waste management challenges:

- **Loss of Economic Value from Non-Hazardous Waste:** Usable fabric scraps and leftover sewing threads are often co-mingled, openly burned, or dumped into municipal landfills despite significant upcycling value.
- **Hazardous Waste Regulatory Compliance Risks:** Chemical dye wastewater, wastewater treatment sludge, and oil-stained cotton rags risk being discarded into domestic waterways due to lack of environmental agency SOP awareness.

TexCycle serves as an intelligent on-device digital assistant capable of sorting textile waste within milliseconds without relying on cellular data or cloud servers.

1.2 Design Philosophy: Minimalist & Industrial-Grade

The application strictly adheres to clean software engineering practices and industrial aesthetics:

- **Zero Chatbot Bloat:** Replaces excessive emojis with crisp, objective, and unambiguous operational terminology.
- **Standard Material Design Iconography:** Standardized visual cues for navigation, performance indicators, and safety notices.
- **One-Hand Ergonomics:** Action triggers and camera viewfinders optimized for fast, repetitive usage on workshop cutting tables.

2. UI ARCHITECTURE, PAGE STRUCTURE, & SYSTEM FEATURES

TexCycle Version 1.2 is structured around 7 operational screens organized in a responsive flat navigation hierarchy:

2.1 Screen 1: Splash Screen & Async Preloader (`SplashView`)

Startup • Parallel Preload

UI COMPONENTS	TECHNICAL FEATURES	USER FLOW
Circular eco-green emblem, application title *TexCycle*, system subtitle, glowing neon green progress bar, async stage status indicator, release tag *v1.2 On-Device.*	Parallel asynchronous initialization (SQLite connection, MobileNetV2 TFLite weight warming in RAM, camera calibration). Zero white screen freeze & smooth 900ms fade transition.	App launches; SplashView appears in <100ms. In the background, AI models and database connections initialize. Progress hits 100% and transitions to HomeView.

2.2 Screen 2: Home & Monitoring Dashboard (`HomeView`)

Dashboard • Early Warning

UI COMPONENTS	TECHNICAL FEATURES	USER FLOW
Offline status header; 30-Day Analytics Card (Total scans, Non-B3 %, B3 %, visual ratio bar); 7-Day Hazardous Accumulation Early Warning Banner; Hero Action CTA *Scan Waste Now.*	Real-time data aggregation directly from local SQLite database without cloud latency; zero text-overflow protection via `FittedBox` & `Expanded`; 7-day environmental risk threshold monitoring.	The artisan reviews recent waste sorting compliance. When new scraps are generated, tapping the prominent green CTA immediately activates the live scanning lens.

2.3 Screen 3: Smart Camera & Advance Lens HUD (`ScanView`)

Computer Vision • Real-Time HUD

UI COMPONENTS	TECHNICAL FEATURES	USER FLOW
Live rounded viewfinder; Real-time lighting status banner; Dynamic reticle focus ring; Scanning radar line; Live Lux gauge (`Lux: XX%`); 1-Tap Auto-Flash chip; 1x/2x digital zoom; 76px shutter button.	Real-time Y-channel luminance sampling at 4 FPS (<1% CPU); inter-frame jitter motion detection; hardware torch mode toggling; physical Android back button interception via `PopScope`.	Camera frames fabric scraps at 30–40cm distance. If dim (<42 Lux), the banner guides flash activation. When steady & focused green, shutter triggers inference in ~68ms.

2.4 Screen 4: Diagnostic Result & Inline DIY Projects (`ResultView`)

Diagnosis • Inline DIY

UI COMPONENTS	TECHNICAL FEATURES	USER FLOW
Memory-optimized photo thumbnail (cacheWidth: 800); Decisive Non-B3 (Green) or B3 (Red) badge; Confidence score %; Market valuation (Rp/kg); Seamless DIY project accordion; 5 DLH SOPs; Weight input form.	3-Tier confidence validation; inline DIY craft recommendations without navigating away; atomic SQLite transactions with automatic image file rollback if storage fails.	Classification renders instantly. For fabric scraps, upcycling patterns (tote bag, pouch) appear alongside resale values. User enters scrap weight (e.g., 2.5 kg) and taps *Save to History*.

2.5 Screen 5: Audit History & CSV Report Export (`HistoryView`)

Audit • CSV Export

UI COMPONENTS	TECHNICAL FEATURES	USER FLOW
Live text search field; Filtering choice chips (All, Non-B3, B3, Verification Required); Shimmer skeleton loading; Downsampled photo list (cacheWidth: 150); Full detail modal sheet; CSV export CTA.	Instant keystroke search filtering; memory downsampling (~90 KB) ensuring zero memory leaks across large datasets; Excel-compatible CSV generator; OS-native share intent via WhatsApp/Email.	The manager reviews weekly waste logs, filters by hazardous types, or taps any card for detailed audit specs. Tapping *Export CSV* generates formal manifests for environmental audits.

2.6 Screen 6: Offline DIY Upcycling Guide (`GuideView`)

Offline Catalog

UI Components: Raw material filter chips (Large, Medium, Small Scraps, Thread, Packaging); Interactive project cards (Tote Bag, Pouch, Scrunchie, Brooch, Tassel); Accordion tools & step-by-step instructions.

Features & Flow: Fully offline embedded guide catalog via `DIYData`. Users select their available scrap dimensions and follow step-by-step procedures to create marketable upcycled goods.

2.7 Screen 7: Settings & Storage Diagnostics (`SettingsView`)

Bilingual • Storage

UI Components: Instant 1-tap bilingual language toggle (ID / EN); Version 1.2 profile badge; Live disk storage breakdown (MB for SQLite & photos); Maintenance CTAs (CSV export, trim history >6 mo, reset); Privacy guarantee.

Features & Flow: Global reactive localization via `AppLocale` & `AppText` without app restarts; real-time disk storage calculation via `path_provider`; safe cache purging.

3. FEATURE SPECIFICATIONS & TECHNICAL INNOVATIONS

Autonomous Edge AI Inference: Machine learning inference executes directly on mobile CPU via C++ TensorFlow Lite JNI bindings. Sub-68ms latency with zero ongoing cloud server costs.

3-Tier Validation & Fluent Localization: Segregates confidence scores into 3 strict certainty zones (Valid, Verify, Reject). Built-in reactive bilingual UI (English & Indonesian) without text bloat.

Atomic Rollback & Zero Orphan Files: Stores compressed 640px JPEG (70% quality) and commits SQLite entry in tandem. If database commit fails, cached image is immediately pruned from disk.

4. SYSTEM FLOWCHARTS

Ultra-HD Standalone Diagram Assets: Available at `TexCycle_Project/diagrams/flowchart_aplikasi_texcycle.png` (2000×1180 Ultra-HD) & `flowchart_aplikasi_texcycle.svg` (Scalable Vector).

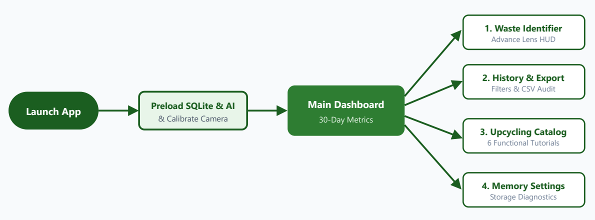

Figure 1: TexCycle Application Lifecycle Flowchart

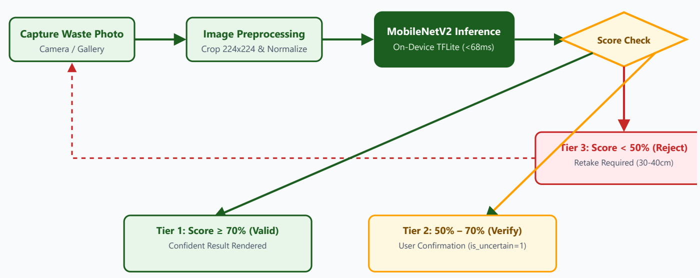

Figure 2: 3-Tier On-Device AI Validation Lifecycle

5. LOCAL SQLITE DATABASE ARCHITECTURE

Ultra-HD Standalone Diagram Assets: Available at `TexCycle_Project/diagrams/skema_database_texcycle.png` (2000×1180 Ultra-HD) & `skema_database_texcycle.svg` (Scalable Vector).

5.1 `scans` Table Schema & Data Dictionary

Column Name	Data Type	Attributes	Description
`id`	`INTEGER`	PK, AUTO	Unique scan history identifier
`image_path`	`TEXT`	NOT NULL	Relative/absolute path of 70% compressed JPEG
`jenis_id`	`TEXT`	NOT NULL	Target waste category ID token
`label_nama`	`TEXT`	NOT NULL	Full descriptive category label for UI
`kategori_b3`	`INTEGER`	INDEX, NOT NULL	Regulatory status: 0 (Non-B3) / 1 (Hazardous B3)
`confidence`	`REAL`	NOT NULL	Model prediction probability score (0.0–1.0)
`is_uncertain`	`INTEGER`	DEFAULT 0	Validation flag: 0 (Valid) / 1 (Needs Review)
`catatan`	`TEXT`	NULLABLE	Weight estimate (kg) / batch / roll identifier
`created_at`	`TEXT`	INDEX, NOT NULL	ISO-8601 compliant audit timestamp

5.2 Indexing Strategy & Data Integrity

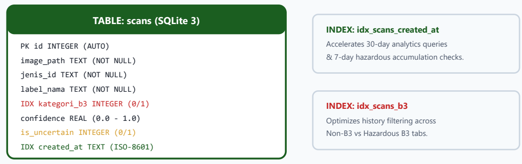

Figure 3: Entity Relationship & Indexing Strategy

Zero Orphan File Protection: Image is written to disk followed by an atomic SQLite commit. If database execution fails, the newly created image file is immediately purged.

6. MACHINE LEARNING MODEL TRAINING PIPELINE

6.1 MobileNetV2 Transfer Learning Architecture

The vision pipeline employs **MobileNetV2** pre-trained on ImageNet with inverted residual bottlenecks:

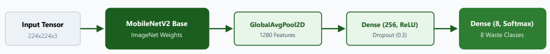

Figure 4: MobileNetV2 Vision Pipeline

TFLite Quantization: Footprint shrinks from 28.4 MB (FP32) to **8.5 MB** (INT8 Dynamic Range, 70.1% reduction) with average inference latency of **68 ms**.

6.2 8-Class Waste Taxonomy & Evaluation Metrics

ID	Waste Category	Status	Precision	Recall	F1-Score
01	Large Fabric Scraps (>30 cm)	NON-B3	0.94	0.92	**0.93**
02	Medium Fabric Scraps (10–30 cm)	NON-B3	0.91	0.93	**0.92**
03	Small Fabric Trimmings (<10 cm)	NON-B3	0.89	0.88	**0.88**
04	Sewing Thread & Bobbin Waste	NON-B3	0.95	0.96	**0.95**
05	Plastic Wrapping & Cardboard Cones	NON-B3	0.93	0.91	**0.92**
06	Chemical Dye Wastewater	HAZARDOUS B3	0.96	0.97	**0.96**
07	Sludge & Effluent Settlings	HAZARDOUS B3	0.97	0.98	**0.97**
08	Contaminated Oil-Stained Rags	HAZARDOUS B3	0.94	0.92	**0.93**

7. APPLICATION FEATURE EVOLUTION (VERSIONS 1.0, 1.1, & 1.2)

Feature / Aspect	Version 1.0 (MVP Baseline)	Version 1.1 (UX & Live Lens)	Version 1.2 (Advance Lens & Security)
Camera Subsystem	Native Device Intent	Live In-App Camera Preview	**Advance Computer Vision Lens** (Y-Luminance & Jitter Sensing)
Optical Sensors & HUD	None (Static)	30–40 cm Distance Text Hint	**Live HUD Lux Gauge**, Motion Jitter Sensor, 1-Tap Flash, 1x/2x Zoom
Startup (Preloader)	White Blank Screen (UI Freeze)	White Blank Screen (UI Freeze)	**SplashView Preloader** (Parallel SQLite & TFLite RAM warming)
Upcycling Recommendations	Separate Manual Menu	Integrated in Scan Result	**Directly Integrated** with collapsible DIY steps accordion
Security & Certification	Debug Development Key	Debug Development Key	**Official Production Keystore** (RSA 2048, SHA-256) Schemes v1–v4
Storage Permissions	Requires WRITE_EXTERNAL_STORAGE	Requires WRITE_EXTERNAL_STORAGE	**Zero Storage Permissions** (Play Protect Clean & Private Sandbox)

Feature / Aspect	Version 1.0 (MVP Baseline)	Version 1.1 (UX & Live Lens)	Version 1.2 (Advance Lens & Security)
APK Package Footprint	~76.8 MB (Universal)	~76.8 MB (Universal)	**~24.6 – 29.1 MB** (Split-per-ABI, 67% Lighter Download)

🥉 Version 1.0 (MVP Baseline)

- Offline 8-class MobileNetV2 classifier
- 3-tier confidence validation engine
- Local SQLite database & CSV export
- 5 Mandatory DLH Environmental SOPs

🥈 Version 1.1 (UX & Live Lens)

- Native in-app camera viewfinder
- Seamless inline DIY upcycling card
- Shimmer skeleton loading placeholders
- Recycled branding & overflow protection

🥇 Version 1.2 (Security & Advance Lens)

- Official production keystore (Anti-Flagging)
- Zero external storage permissions
- SplashView preloader (Zero white freeze)
- Y-Luminance, Jitter HUD, & Split APK ~25MB

8. CONCLUSION & IMPLEMENTATION READINESS

The **TexCycle** application demonstrates that on-device machine learning (*Edge AI*) can be practically deployed in micro-scale garment enterprises with zero recurring server costs. The system guarantees regulatory environmental compliance, safeguards artisan data privacy, and creates tangible economic value by converting discarded textile waste into marketable, high-utility upcycled products.